

```
import math
```

```
def calculate_co_amoxiclav(total_dose, total_day, times_per_day):
```

```
    """
```

```
    Calculate dosage for co-amoxiclav suspension formulations
```

```
    """
```

```
    # Calculate for 643mg/5ml formulation
```

```
    n1_susp643 = math.ceil((total_dose * 5 / 643) * total_day / 75)
```

```
    n2_susp643 = math.ceil((total_dose * 5 / 643) * total_day / 100)
```

```
    q_susp643 = math.ceil((total_dose * 5 / 643) / times_per_day)
```

```
    print(f"\nCo-Amoxiclav 643mg/5ml:")
```

```
    print(f"- Dosage: {q_susp643}ml every {24/times_perday} hours")
```

```
    print(f"- Total treatment duration: {total_day} days")
```

```
    print(f"- Bottles needed: {n1_susp643} (75ml) OR {n2_susp643} (100ml)")
```

```
    # Calculate for 457mg/5ml formulation
```

```
    n1_susp457 = math.ceil((total_dose * 5 / 457) * total_day / 75)
```

```
    n2_susp457 = math.ceil((total_dose * 5 / 457) * total_day / 100)
```

```
    q_susp457 = math.ceil((total_dose * 5 / 457) / times_per_day)
```

```
    print(f"\nCo-Amoxiclav 457mg/5ml:")
```

```
    print(f"- Dosage: {q_susp457}ml every {24/times_perday} hours")
```

```
    print(f"- Total treatment duration: {total_day} days")
```

```
    print(f"- Bottles needed: {n1_susp457} (75ml) OR {n2_susp457} (100ml)")
```

```
def calculate_amoxicillin(total_dose, total_day, times_per_day):
```

```
    """
```

```
    Calculate dosage for amoxicillin suspension formulations
```

```
    """
```

```
    # Calculate for 400mg/5ml formulation
```

```
    n_faramox400 = math.ceil((total_dose * 5 / 400) * total_day / 70)
```

```
    q_faramox400 = math.ceil((total_dose * 5 / 400) / times_per_day)
```

```
    print(f"\nAmoxicillin 400mg/5ml:")
```

```
    print(f"- Dosage: {q_faramox400}ml every {24/times_perday} hours")
```

```
    print(f"- Total treatment duration: {total_day} days")
```

```
    print(f"- Bottles needed: {n_faramox400} (70ml)")
```

```
    # Calculate for 200mg/5ml formulation
```

```
    n_faramox200 = math.ceil((total_dose * 5 / 200) * total_day / 70)
```

```
    q_faramox200 = math.ceil((total_dose * 5 / 200) / times_per_day)
```

```
    print(f"\nAmoxicillin 200mg/5ml:")
```

```
    print(f"- Dosage: {q_faramox200}ml every {24/times_perday} hours")
```

```
    print(f"- Total treatment duration: {total_day} days")
```

```
    print(f"- Bottles needed: {n_faramox200} (70ml)")
```

```
def sinusitis_treatment():
```

```
"""
```

Main function to determine sinusitis treatment

```
"""
```

```
print("\n=== Sinusitis Treatment Calculator ===")
```

```
print("Please enter patient details:\n")
```

```
try:
```

```
    # Get patient information
```

```
    weight = float(input("Patient weight (kg): "))
```

```
    age = int(input("Patient age (years): "))
```

```
    temperature = float(input("Temperature (°C): "))
```

```
    pain = input("Pain level (mild/moderate/severe): ").lower()
```

```
    # Get treatment frequency
```

```
    times_per_day_input = input("\nEnter frequency (2 or 3), or 'default' for default  
frequency: ").lower()
```

```
    if times_per_day_input == 'default':
```

```
        times_per_day = 2
```

```
    else:
```

```
        times_per_day = int(times_per_day_input)
```

```
    # Determine severity
```

```
    severity = input("\nEnter 'high_risk' if any of:\n- Age <2 years\n- Antibiotics in last 3  
months\n- No response to amoxicillin in 3 days\nPerform *Moderate/Severe  
illness*\nOtherwise enter 'low_risk': ").lower()
```

```
    # Calculate total daily dose (90mg/kg/day, max 3000mg/day)
```

```
    total_dose = 90 * weight
```

```
    if total_dose > 3000:
```

```
        total_dose = 3000
```

```
    # Get treatment duration
```

```
    total_day_input = input("\nEnter treatment duration (10-14), or 'default' for 12 days:  
").lower()
```

```
    if total_day_input == 'default':
```

```
        total_day = 12
```

```
    else:
```

```
        total_day = int(total_day_input)
```

```
    # Determine appropriate antibiotic
```

```
    print("\n=== Treatment Recommendation ===")
```

```
    if severity == 'high_risk':
```

```
        print("\nHigh risk patient - Co-Amoxiclav recommended:")
```

```
        calculate_co_amoxiclav(total_dose, total_day, times_per_day)
```

else:

```
print("\nLow risk patient - Amoxicillin recommended:")  
calculate_amoxicillin(total_dose, total_day, times_per_day)
```

Print important notes

```
print("\n=== Important Notes ===")  
print("- Complete full course of antibiotics")  
print("- Symptoms should improve within 48-72 hours")  
print("- Return if symptoms worsen or persist beyond 10 days")
```

Disclaimer

```
print("\n=== DISCLAIMER ===")  
print("This tool is for educational purposes only. It does not replace clinical judgment.")  
print("Always consult with a physician for medical decisions.")
```

except ValueError:

```
print("\nError: Please enter valid numeric values for weight, age and temperature.")
```

Run the program

```
if __name__ == "__main__":  
    sinusitis_treatment()
```